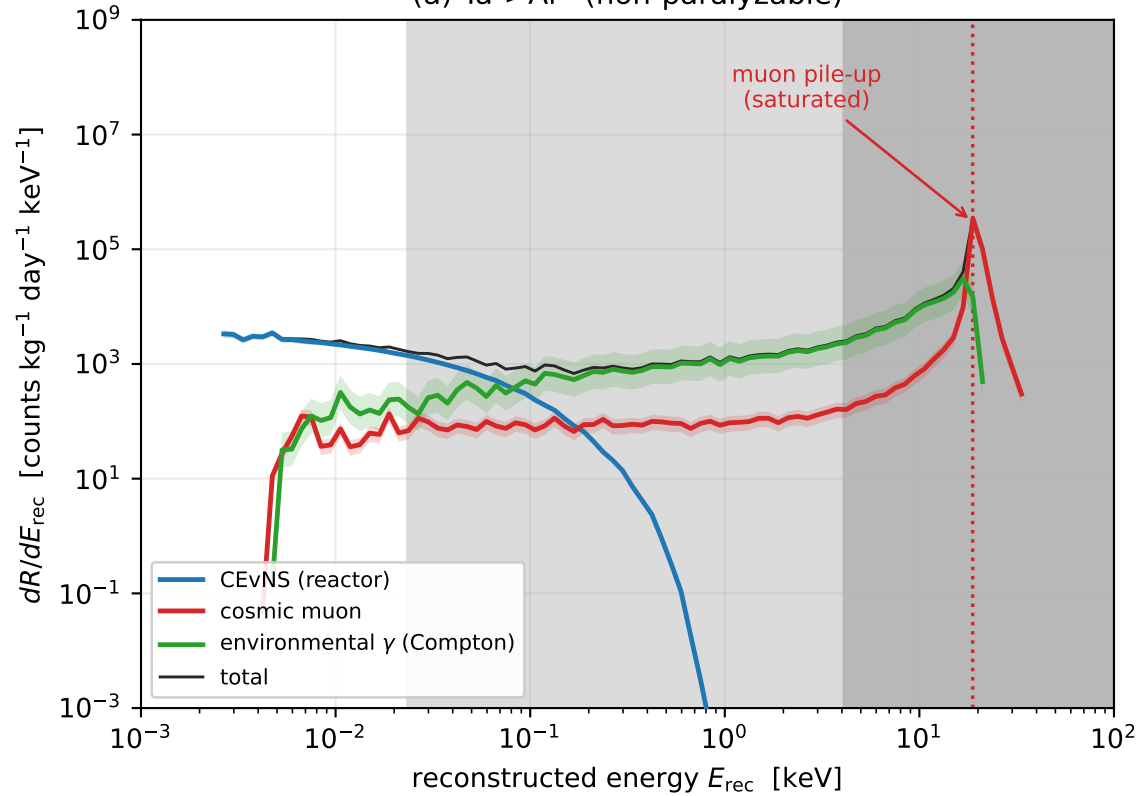
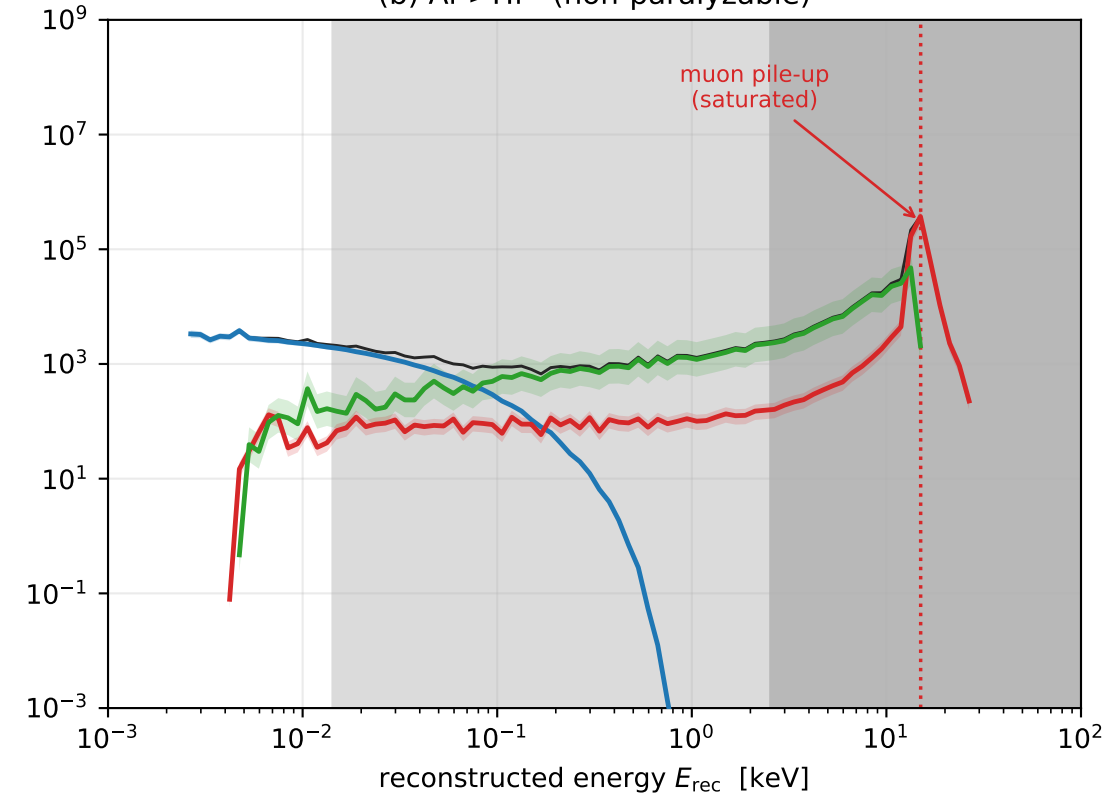


Reconstructed-energy differential-rate spectra — QPD Ge wafer (4'' × 4'' × 2 mm, ~110 g, per-kg), stage-1 deliverable

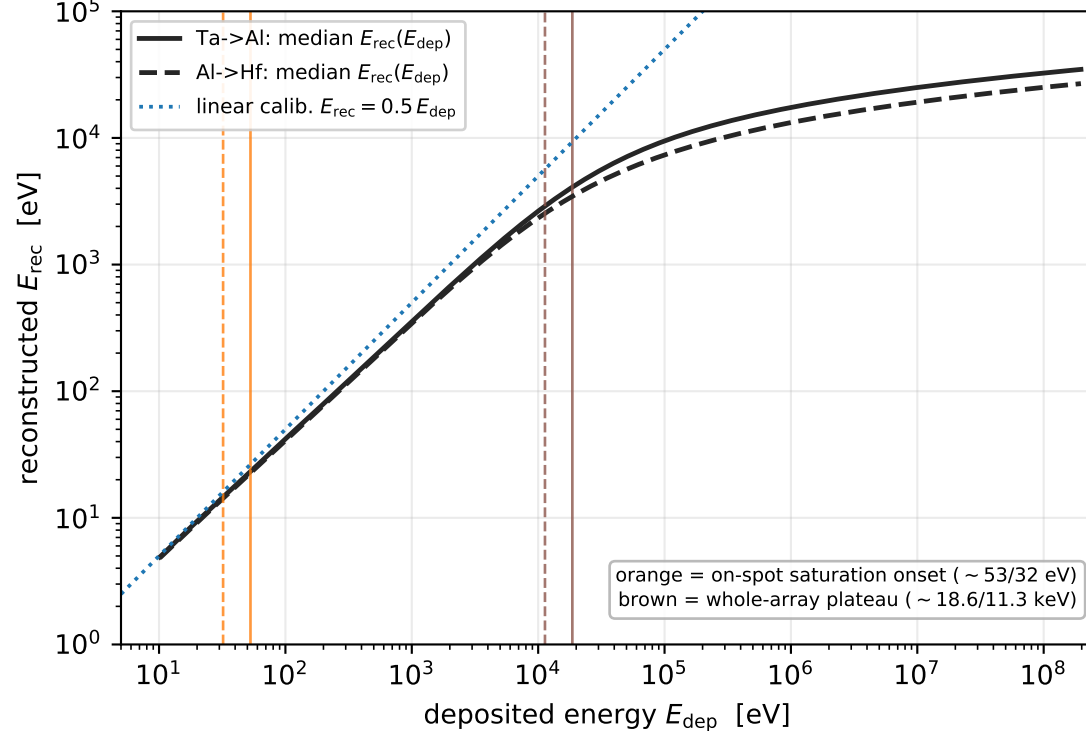
(a) Ta->Al (non-paralyzable)



(b) Al->Hf (non-paralyzable)



(c) true → reconstructed mapping (Phase-5 non-paralyzable response; reused)



Honest caveats (stage-1):

- Axis is RECONSTRUCTED energy E_{rec} , not deposited.
- NO literature anchor for the saturated-regime shape; validation is limiting-cases-only. The muon channel is reconstructed ENTIRELY in saturation — its tens-of-keV pile-up is an instrument artifact of the modelled ceiling, NOT a physical spectral line.
- CEvNS reconstructs to TENS OF eV E_{rec} (peak ~42 eV, ~85% below 100 eV); the flagship signal sits very low.
- Bands: CEvNS 1 σ reactor-flux; Compton factor-2 site-dependent γ flux; muon $\pm 30\%$ normalization.
- Grey shading = deposits in saturation ($E_{\text{dep}} > \text{onset}$); darker = whole-array plateau. Non-paralyzable is the deliverable (CONVENTIONS §F resolved 2026-07-21).

saturation delimitation

- saturation region ($E_{\text{dep}} > \text{onset}$)
- whole-array plateau region
- muon E_{rec} pile-up